

MID term Syllabus

Introduction to image processing: Overview and significance of image processing, Components of an image processing system, Image formation principles, Digitization of images, Continuous and discrete image representations,

Basic image processing operations: Neighbours of pixel, Adjacency and connectivity, Regions and boundaries, Distance measures, Arithmetic operations, Logical operations, Geometrical operations, Convolution and correlation operations,

Image pre-processing techniques and feature extraction: Noise reduction and filtering, Smoothing and sharpening, Histogram equalization, Image enhancement techniques, Point detection, Line and edge detection, Colour, texture, shape and structure features, Spatial and frequency domain features, Corner detection, Hough transform